KILNDECK · DESIGN DECISION RECORD

KilnDeck TRL 2 review decisions

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DOCUMENT CONTROL

Document ID	KND-DDR-001	Version	0.1
Status	Draft	Date	2026-10-03
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/kilndeck		
Source	docs/decisions/0001-trl2-review-decisions.md @ 9f92bd8		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-03	Amish Chadha	TRL 2 review items decided under Amish's 2026-10-03 pre-approval

0001: TRL 2 review decisions

- **Date:** 2026-10-03
- **Status:** decided. Decided by Amish under his pre-approval of 2026-10-03: "start with the first 14 repos from the list of 29 projects. I pre-approve the batch runs along with any recommendations you come up with. I also accept any cost overruns or variations from the assumed scope cost."

Context

The scaffold (KND-PRB-001, KND-PRC-001 and KND-REQ-001, all v0.1) left five open questions and described the deck only in outline: "a pair of rails runs along the top of the kiln, carried on supports that bear on the kiln's structural walls", with a small deck that the fireman pushes from hole to hole. Populating the concept to TRL 2 meant settling how the deck reaches every feed hole while bearing only on the walls. Amish pre-approved every recommendation in this batch, so each item below is decided, not proposed. Items that touch safety take the conservative option and say what evidence would relax it. Partners and regions are the first candidates to approach, not agreements.

Options considered

Table 1. Options.

#	Item	Options
D1	How the deck reaches every hole	(a) small deck on rails laid along the crust on cross beams; (b) deck that spans the trench wall to wall and rolls along the kiln on one track per wall top; (c) fixed bridges moved by hand
D2	Track layout	(a) full circuit round the kiln; (b) movable sections on the straight sides, leapfrogged ahead of the fire
D3	How fuel reaches the holes	(a) hopper on the deck floor with a chute through the floor; (b) hopper on a carriage that runs along the outside of one side truss and feeds the row beside the deck
D4	How the deck is moved	(a) pushed from the wall top at one end; (b) hand wheel on the deck driving one wheel at each end through a line shaft
D5	Guardrail height	(a) 1.0 m as in the scaffold R2; (b) 1.1 m top rail with knee rail and toe board, after EN ISO 14122-3
D6	Floor	(a) plain steel chequer plate; (b) insulated panel: steel top, calcium silicate board, reflective aluminium underside
D7	Fuels	(a) coal only; (b) coal and chopped biomass, with the dose set by the number of pockets
D8	Co-design partner	Clean-kiln programmes and kiln owners' associations in Pakistan, India, Bangladesh and Nepal
D9	Budget, pitch and problem	Keep or change

Decision

Table 2. Items decided on 2026-10-03.

#	Item	Decision	Why
D1	Reaching every hole	(b): the deck spans the trench and rolls along the kiln on two wall-top tracks. It is still the railed rolling deck of the pitch; it now stands only on the walls	Option (a) puts track supports on the crust or needs cross beams across the trench anyway; (b) bears on the walls alone and reaches every hole in the trench by moving along the kiln
D2	Track layout	(b): 3 m track modules on the straight sides, laid ahead of the fire and lifted from behind it (the fire moves 6 to 10 m a day); the curved ends are not served by this prototype	Lighter, cheaper and needs no curved track; a full circuit is a later option
D3	Fuel delivery	(b): hopper carriage on side B feeding the row of holes 1.2 m from the deck centre line	Keeps the floor whole (no slot over the fire) and keeps the fireman's feet at least 0.6 m from an open hole
D4	Moving the deck	(b): hand wheel and line shaft driving one wheel at each end, so both ends move together and the 6.6 m deck cannot skew and jam	Pushing one end of a long deck on a short wheelbase risks skewing; the fireman also stays on the deck
D5	Guardrail	(b): top rail 1.1 m above the floor, knee rail at 0.55 m, toe board 0.15 m; R2 restated	Conservative: this is a fall-prevention barrier over a fire. No evidence is expected to relax it; the higher rail costs almost nothing
D6	Floor	(b): insulated panels with a reflective underside	The crust is estimated at 150 °C; plain plate would pass most of that to the feet
D7	Fuels	(b): both. The rotor gives about 0.49 kg of coal or about 0.15 kg of biomass per pocket (estimates); the fireman sets the dose by counting pockets	Kilns switch fuels; one rotor that works with both keeps the design simple
D8	Co-design partner	First candidate to approach: a clean-kiln research programme in Bangladesh already running fuel-feeding training (for example the Stanford and BUET kiln team). Second: a kiln owners' association in Punjab, Pakistan. Neither has been approached	Training in continuous feeding is already in place there, and the fall hazard is documented in Punjab
D9	Budget, pitch and problem	No change. `budget_usd` stays at USD 3,500 as the value-engineering target	The constructable design is estimated at USD 2,627

Consequences

- KND-PRB-001, KND-PRC-001 and KND-REQ-001 are revised to v0.2. R2 is restated (1.1 m top rail, knee rail, toe board), R4 is restated as the force on the hand wheel rim, and R12 (no piece over 40 kg) and R13 (the deck cannot run off its track) are added.
- The open questions of KND-PRB-001 are answered in the precis: sections, not a full circuit (D2); rail heat handled by sliding fishplate joints and a floating inner wheel (KND-DDR-002); coal and biomass (D7). What firemen think of the deck stays open for the co-design partner, as research rather than a design decision.
- The kiln dimensions in the calculation note (trench width, wall-top width, feed hole grid, crust temperature) are assumptions until a survey of the partner kiln.
- TRL 4 is on hold by the portfolio phase. Nothing in this record authorises building or testing.